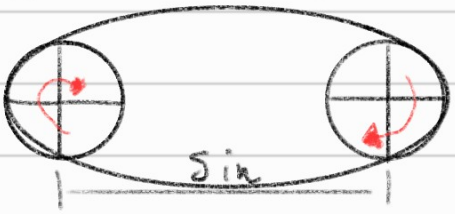
Belt design



Knowns.

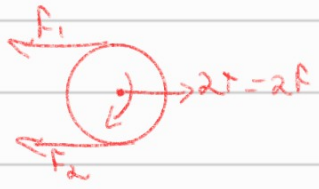
Parallel axes seperated by a distance of 4-6 inches

- Pulley diameter: 3 inches $= d_p$
- Rot. speed: 1000-2000 RPM → Chose: 2000 RPM $= \omega$
- Belt tension: 4-6 lbf → Chose: 6 lbf $= T$
- Pulley material → Steel
- S F (Fos) = 3
- Belt Material → Polyurethane rubber
  - σ_y = 3.25 MPa
  - Strain at 20%

Unknowns

F_1, F_2, d_{belt}

Assuming $\mu = 1$ (No slippage)



$\Sigma F_x = -F_1 - F_2 + 2F = 0$

$2F = F_1 + F_2$

$\frac{F_1}{F_2} = e^{\mu\phi}$ → $\frac{2F - F_2}{F_2} = e^{\mu\phi}$

$\frac{2F}{F_2} = 1 + e^{\mu\phi}$

$F_1 = 2(T) - F_2$

$F_1 = 11.5029$ lbf

$F_2 = \frac{2F}{1+e^{\mu\phi}} = \frac{2(6\ lbf)}{1+e^{(1)(\pi)}}$

$F_2 = 0.4971$ lbf

$\sigma = \frac{S_y}{S.F.} = \frac{F}{A}$

$A_b = \frac{F_1 \cdot S.F.}{S_y}$

$\frac{\pi}{4} d_b^2 = \frac{F_1 \cdot (S.F.)}{S_y}$

$d_b = \sqrt{\frac{4 \cdot F_1 \cdot (S.F.)}{S_y \cdot \pi}}$

$d_b = 0.30542804$ in → Closest industry belt size?

$A_b = \frac{\pi}{4} d_b^2 = 0.073267\ in^2$

$5/16" = 0.3125$ in

Belt part # 3044K707

Belt circumference

$C = 2L + \pi(d_p + d_b)$

$C = 2(6) + \pi(3 + 0.30542804)$

$C = 22.384$ in

Shaft Design

Using 1060 hardened steel → $\sigma_y = 98200\ psi = S_y$

Knowns:	Unknowns
$T = F = 6\ lbf$	τ, d_{min} of shaft
S.F. = 3	
$\varnothing_{Pulley} = 3''$ $= d_p$	
Pulley material: steel	
$F_1 = 11.5024\ lbf$	
$F_2 = 0.4971\ lbf$	
$d_{belt} = 0.305429804''$	
$S_y = 3.25\ Mpa = 471\ psi$ at 20% min	
Rot. speed = 2000 RPM $= \omega$	
$\mu =$	



$T = (F_1 - F_2)\frac{d_p}{2} = (F_1 - F_2)(1.5)$

$T = 16.50874204\ 2\ lbf \cdot in$

$V_{max}(shear_{max}) = 65000\ psi$

$\tau_{max} = \frac{V_{max}}{S.F} = 16250\ psi$ — used website for value

$\tau_{max} = \frac{16T}{\pi d^3}$

$d = \sqrt[3]{16T/\pi\tau_{max}}$

$d = 0.172959632$

$d_s = 0.1730\ in$ — Min diameter of shaft

https://www.makeitfrom.com/material-properties/SAE-AISI-1060-S58C-G10600-Carbon-Steel

Journal Bearings

Knowns:

$T = F = 6$ lbf

S.F. = 3

$\varnothing_{Pulley} = 3'' = d_p$

Pulley material: steel

$F_1 = 11.5029$ lbf

$F_2 = 0.4971$ lbf

$d_{belt} = 0.30542804$ in

$S_y = 3.25$ Mpa $= 471$ Psi at 20°C

Rot. speed $= 2000$ RPM $= \omega$

$\nu = 1$

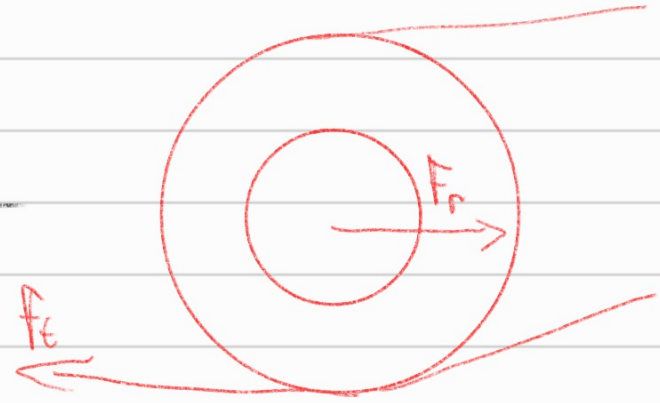
$d_{shaft} = 0.1730$ in

$\tau_{shaft} = 16250$ lb·in

$\sigma_{shaft} = 98200$ Psi

Unknowns

f_t, f_r, P, V



$$f_t = \frac{2T}{d_s} = 64.38054 \text{ lb}$$

$$f_r = f_t \tan\alpha = F_t \tan(20^\circ) = 25.2525 \text{ lb}$$

$$P = \frac{F}{A} = \frac{f_r}{L \cdot d} = \frac{F_r}{3\text{in} \cdot (d_{shaft})} = 48.6674632 \text{ Psi}$$

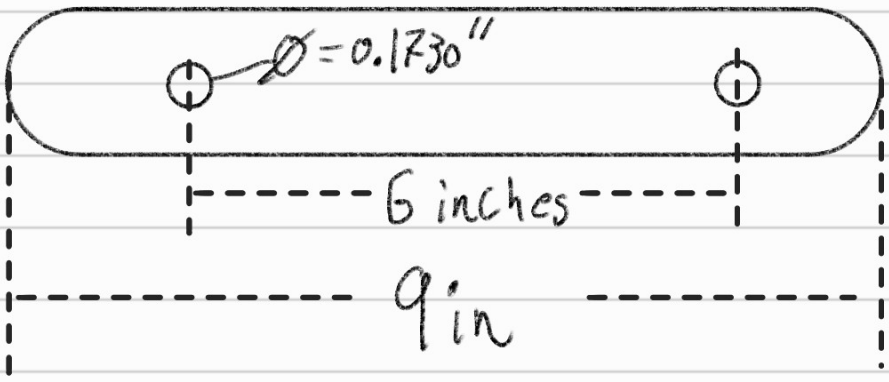
$$V = \omega \pi d_{shaft} = 1{,}086.7345 \text{ ft/min}$$

$$PV = 52{,}88.61 \text{ psi} \cdot \text{ft/min} = PV$$

lubricated Graphite is the only material that will withstand the PV of 50,000 Plus.

Bearing Material	Load Capacity (psi)	Maximum Temperature (°F)	Surface Speed, V_{max} (max. fpm)	PV Limit P = psi load V = surface ft/min
Acetal	1000	180	1000	3000
Graphite (dry)	600	750	2500	15,000
Graphite (lubricated)	600	750	2500	150,000
Nylon, Polycarbonate	1000	200	1000	3000
Nylon composite	...	400	...	16,000
Phenolics	6000	200	2500	15,000
Porous bronze	4500	160	1500	50,000
Porous iron	8000	160	800	50,000
Porous metals	4000–8000	150	1500	50,000
Virgin Teflon (TFE)	500	500	50	1000
Reinforced Teflon	2500	500	1000	10,000–15,000
TFE fabric	60,000	500	150	25,000
Rubber	50	150	4000	15,000
Maple & Lignum Vitae	2000	150	2000	15,000

Frame Design

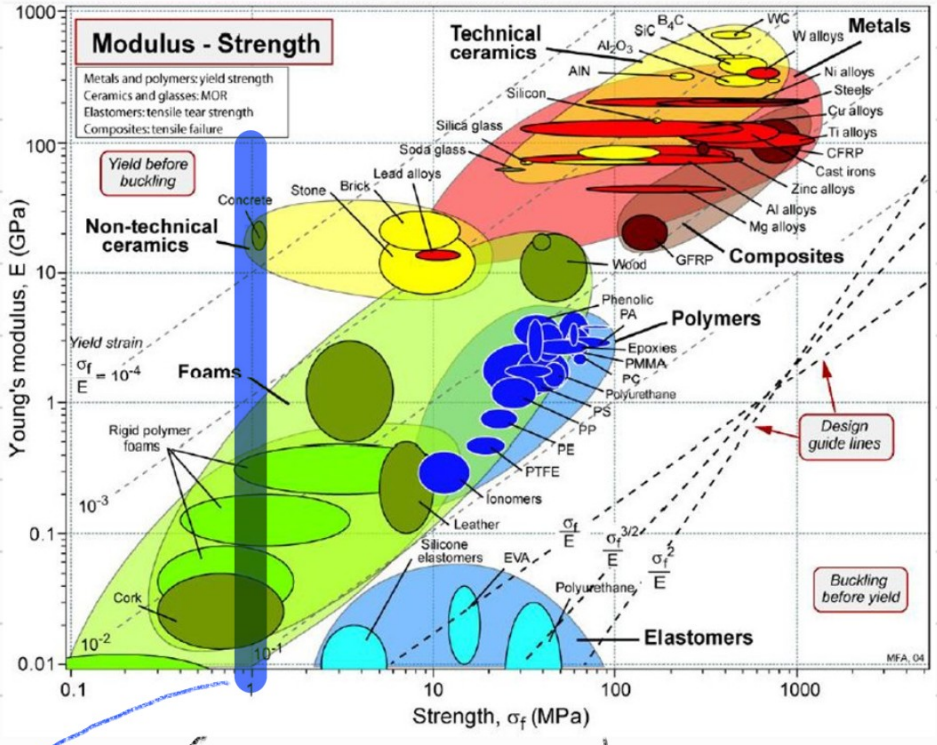


48.6674632 psi = 0.33555 MPa

S.F. x Pressure = 3 x 0.33555 MPa = 1.00665 MPa

Using Ashby plots to simplify the material selection process.

Stronger



lighter

located around the rigid foams, should withstand the pressure the bearings will encounter.